

Fermi–Amaldi-Based Local Mixing Function

Ivan P. Bosko and Viktor N. Staroverov

Department of Chemistry, The University of Western Ontario, London, Ontario N6A 5B7, Canada

(Dated: March 5, 2026)

I. MAIN TEXT

Here we propose a new local mixing function (LMF) for local hybrid functionals. This LMF is Fermi–Amaldi (FA) based and represents the ratio between the FA and the exact exchange energy densities. The first is given by

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (1)$$

and the latter is expressed via molecular orbitals $\{\varphi_i(\mathbf{r})\}$ and the ground-state density matrix $\mathbf{D}$ as

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{2} \sum_{pqrs} D_{pq} D_{rs} \int \frac{\varphi_p^*(\mathbf{r}) \varphi_s(\mathbf{r}) \varphi_r^*(\mathbf{r}') \varphi_q(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (2)$$

We can show that the proposed LMF satisfies the zero

to one boundary range:

$$e_x^{\text{exact}} \leq 0,$$

$$e_x^{\text{FA}} \leq 0,$$

$$e_x^{\text{exact}} \leq e_x^{\text{FA}},$$

thus

$$0 \leq \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} \leq 1. \quad (3)$$

For one-electron regions, $e_x^{\text{FA}} = e_x^{\text{exact}}$, therefore, the one-electron constraint is satisfied. A particular case of a one-electron limit, i.e., the asymptotic limit ($r \rightarrow \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1 \quad (4)$$

since both energy densities approach $-1/2r$ as $r \rightarrow \infty$.